



Tolerance development in *Listeria monocytogenes*-*Escherichia coli* dual-species biofilms after sublethal exposures to pronase-benzalkonium chloride combined treatments

Pedro Rodríguez-López, Marta López Cabo*

Department of Microbiology and Technology of Marine Products, Instituto de Investigaciones Marinas (IIM-CSIC), Eduardo Cabello 6, 36208 Vigo, Pontevedra, Spain



ARTICLE INFO

Article history:

Received 27 December 2016

Received in revised form

2 May 2017

Accepted 1 June 2017

Available online 1 June 2017

Keywords:

Benzalkonium chloride

Biofilm

Enzyme

Escherichia coli

Fluorescence microscopy

Listeria monocytogenes

Pronase

Tolerance

ABSTRACT

This study was designed to assess the effects that sublethal exposures to pronase (PRN) and benzalkonium chloride (BAC) combined treatments have on *Listeria monocytogenes*-*Escherichia coli* dual-species biofilms grown on stainless steel in terms of tolerance development (TD) to these compounds. Additionally, fluorescence microscopy was used to observe the changes of the biofilm structure. PRN-BAC exposure was carried out using three different approaches and TD was evaluated treating biofilms with a final 100 µg/ml PRN followed by 50 µg/ml BAC combined treatment. Results showed that exposure to PRN-BAC significantly decreased the number of adhered *L. monocytogenes* ($P < 0.05$), while *E. coli* counts remained generally unaltered. It was also demonstrated that the incorporation of recovery periods during sublethal exposures increased the tolerance of both species of the mixed biofilm to the final PRN-BAC treatment. Moreover, control biofilms became more resistant to PRN-BAC if longer incubation periods were used. Regardless of the treatment used, log reduction values were generally lower in *L. monocytogenes* compared to *E. coli*. Additionally, microscopy images showed an altered morphology produced by sublethal PRN-BAC in exposed *L. monocytogenes*-*E. coli* dual-species biofilms compared to control samples. Results also demonstrated that *L. monocytogenes*-*E. coli* dual-species biofilms are able to develop tolerance to PRN-BAC combined treatments depending on way they have been previously exposed. Moreover, they suggest that the generation of bacterial tolerance should be included as a parameter for sanitation procedures design.

© 2017 Elsevier Ltd. All rights reserved.

1. Introduction

Listeria monocytogenes and *Escherichia coli* are foodborne pathogens currently considered as an issue of major concern in food safety due to their incidence and morbidity and mortality rates among the population (EFSA, 2015). In food related environments, these pathogens are normally present adhered to surfaces forming part of complex multispecies biofilms (Rodríguez-López et al., 2015). These associations give rise to the establishment of inter-species interactions that can provoke an increased resistance to chemical disinfectants (Holah et al., 2002; Saá Ibusquiza et al., 2012a; van der Veen and Abee, 2011).

In biofilms, such resistance has been proved to be of a

multifactorial nature, however, one of the main reasons is the presence of a robust extracellular polymeric matrix (Abdallah et al., 2014). This structure, constituted by proteins, polysaccharides and extracellular DNA (eDNA) (Flemming and Wingender, 2010), acts as a protective permeability barrier hindering antimicrobials diffusion (Abdallah et al., 2014) and therefore creating a bactericidal concentration gradient into the biofilm (Allison, 2003; Flemming et al., 2007). Hence, cells especially those located in the deepest layers of the biofilms, would be exposed to sublethal concentrations of antimicrobials that would provoke an environmental pressure on the biofilm population and the subsequent selection for bactericide-tolerant phenotypes (Abdallah et al., 2014).

Enzymatic biofilm matrix breakdown previous to disinfectant application is nowadays considered an environmentally friendly alternative for both pathogen control and reduction of the amount of chemicals used. An extensive review on the application of enzymes in the food industry has been recently published (Meireles

* Corresponding author.

E-mail address: marta@iim.csic.es (M.L. Cabo).

et al., 2016).

In particular, hydrolytic enzymes have been proposed as efficient agents to remove biofilms (Augustin et al., 2004; Orgaz et al., 2007; Simões et al., 2010). Especially in mature biofilms, they can be useful to remove from the surface the matrix of biofilms and to facilitate penetration of cleaning and disinfection agents (Abdallah et al., 2014). Among them, proteolytic enzymes have been proved to be effective against biofilms formed by pathogenic bacteria present in food industry premises (Augustin et al., 2004; Molobela et al., 2010).

In this line of research, previous studies carried out in our laboratory have demonstrated that the combined application of pronase (PRN) with benzalkonium chloride (BAC) is able to remove dual species biofilms of *L. monocytogenes* formed on stainless steel (Rodríguez-López et al., 2017).

Tolerance development (TD) after sublethal exposure to BAC and other quaternary ammonium compounds (QACs) has extensively reported in planktonic cells of *L. monocytogenes* (Jiang et al., 2016; Romanova et al., 2006; Saá Ibusquiza et al., 2012b; To et al., 2002) and *E. coli* (Bore et al., 2007; Moen et al., 2012). However, a significantly smaller number of studies involving TD after BAC sublethal exposure in mixed-species biofilms is currently found in the literature and, to the best of authors' knowledge, no studies were found assessing the tolerance to enzyme-BAC combined strategies. The only evidence found to BAC tolerance development in mixed-species biofilms was the study performed by Machado et al. (2012). They demonstrated that after five sublethal exposures to BAC carried out every 24 h, *Pseudomonas aeruginosa*-*E. coli* biofilms grown in polystyrene presented a higher biomass content compared with unexposed samples. Nevertheless, the number of adhered cells was similar in both series.

One of the main advantages of applying enzymes is that they do not pose a selective pressure on bacteria (Meireles et al., 2016). However, if combined with biocides such as BAC, it seems to be important to consider both agents for a correct TD evaluation.

The main aim of this work was to assess the TD to PRN-BAC treatments after sublethal exposure to these compounds in *L. monocytogenes*-*E. coli* dual-species biofilms grown on SS coupons. The exposures were carried out following three different approaches based on the concentrations of the antimicrobials used, the intervals of dosage and the duration of the whole cycle. These approaches intend to mimic different conditions of exposures to the combined enzyme-BAC treatment that could occur in food industries with different programmes of hygienic control.

2. Materials and methods

2.1. Bacterial strains

Dual-species biofilms were formed by *L. monocytogenes* A1 and *Escherichia coli* A14, both isolated in a previous survey performed among pre-sanitised surfaces in a fish processing plant (Rodríguez-López et al., 2015).

Stock cultures of all strains were maintained at -80°C in brain-heart infusion broth (BHI; Biolife, Milan, Italy) containing 50% glycerol 1:1 (v/v) mixed. Working cultures were kept at -20°C in Trypticase Soy Broth (TSB, Cultimed, Barcelona) containing 50% glycerol 1:1 (v/v) mixed.

2.2. Set-up of dual-species biofilms

A 100 μl of working cultures of *L. monocytogenes* or *E. coli* were cultured overnight in 5 ml sterile TSB at 37°C and subcultured and incubated overnight in order to ensure a proper growth.

Inocula preparation was performed as described previously (Rodríguez-López et al., 2017). Briefly, Abs₇₀₀ of each culture was

adjusted to 0.1 ± 0.001 in sterile phosphate buffer saline (PBS) using a 3000 Series scanning spectrophotometer (Cecil Instruments, Cambridge, England) corresponding to a bacterial concentration of about 10^8 CFU/ml. Adjusted cultures were $1:10^4$ diluted in sterile mTSB (TSB supplemented with 2.5 g/l glucose (Vorquímica, S.L., Vigo, Spain) and 0.6% yeast extract (Cultimed, Barcelona, Spain)) to obtain a final concentration of about 10^4 CFU/ml. Then, equal volumes of these adjusted cultures were mixed to obtain the inoculum for dual-species biofilms.

Biofilms were grown on $10 \times 10 \times 1$ mm AISI 316 stainless steel (SS) coupons (Comevisa, Vigo, Spain). Coupon pre-treatment included individual washing with industrial soap (Sutter Wash, Sutter Ibérica, S.A., Madrid, Spain), rinsing with tap water, a final rinse with deionised water and autoclaving at 121°C for 20 min. After this, coupons were individually placed into a 24 flat-bottomed well plate and each well was inoculated with 1 ml of the inoculum. Plates were incubated in a humidified atmosphere at 25°C statically for 2 h for initial adhesion and then in constant shaking at 100 rpm at 25°C until 24 h.

Before any assay was performed, samples (SS coupons) were aseptically collected and briefly immersed in 1 ml sterile PBS to remove loosely attached cells.

2.3. Adhered viable culturable cells (AVC) quantification

Since in studies dealing with antimicrobial testing against biofilms, the quantification of biocidal activities is mostly performed by quantifying the number of adhered viable and culturable cells (AVC) after a given treatment, in this work AVC was referred to this particular cellular pool.

After PBS washing, AVC were collected from three different samples (SS coupons). The surface of each sample was thoroughly swabbed using two sterile cotton swabs moistened in buffered peptone water (BPW, Cultimed, Barcelona). In a previous assay, absence of cells after swabbing was determined staining swabbed samples with Syto9 (Life Technologies, Eugene, OR, USA) and visualised under a Leica DM 6000 epifluorescence microscope (Leica, Wetzlar, Germany) (data not shown). Swabs belonging to the same sample were pooled together in 2 ml of BPW and vortexed vigorously for 1 min in order to release cells. Suspensions were serially diluted in BPW and spread onto agar plates. *Listeria*-PAL-CAM (Liofilchem, Roseto degli Abruzzi, Italy) was used for *L. monocytogenes* selection whereas HiCrome™ Coliform Agar (Sigma-Aldrich, St Louis, MO, USA) supplemented with 5 $\mu\text{g}/\text{ml}$ of vancomycin and cefsulodine (Sigma-Aldrich) was used for *E. coli*. Plates were incubated at 37°C for 24–48 h and results were expressed as the mean in log CFU/cm² or log CFU/cm² reduction depending on the assay. The accepted limit of detection was 25 CFU in the plate of the lowest dilution corresponding to 1.70 log CFU/cm² (Sutton, 2011).

2.4. Microscopy assays

In all cases, for microscopy assays, three different samples (SS coupons) were used. After PBS washing, coupons were stained using LIVE/DEAD® BacLight™ bacterial viability kit (Life Technologies, Eugene, OR, USA) that allows distinguishing total cells with undamaged membranes (green fluorescence) and damaged membranes (red fluorescence). Staining solution was prepared mixing 0.25 μl of propidium iodide and 0.75 μl of Syto9 in 1 ml of filter sterilised distilled water. Fifty microlitres of this solution was poured onto each coupon for 15 min contact time. Subsequently, staining solution was carefully removed and samples were washed three times by immersion in 1 ml of sterile milliQ water for 30 s. Coupons were then air dried and visualised under a Leica DM 6000

epifluorescence microscope using a 40× objective and 10× ocular lenses. In each sample, a randomly chosen field was considered as start point to acquire 12 bit images using a Leica DFC365 FX camera. Each image set was composed by three 5 × 5 mosaics each one covering a total surface of 1.92 × 10⁶ μm². Image analysis was then performed using the Integrated Morphometry Analysis (IMA) module of Metamorph MMAF software (Molecular Devices, Sunnyvale, CA, USA) in order to determine the occupied area (OA) by undamaged (green) cells. Results were expressed in mm².

2.5. Enzymes, BAC and neutralising solutions preparation

Pronase (PRN, from *Streptomyces griseus*, Roche, Mannheim, Germany) stock solution was prepared at concentration 2000 μg/ml dissolved in 0.1 M Tris-HCl (Sigma Aldrich) buffer at pH = 7.5 ± 0.2. After preparation, stock solution was filter sterilised through a 0.2 μm pore diameter syringe filter (Sartorius, Göttingen, Germany) and kept at −20 °C until use.

Benzalkonium chloride (BAC, Guinama, Alboraya, Spain) stock solution was prepared at concentration 4000 μg/ml diluting the commercial solution in sterile distilled water. The solution was kept at 4 °C until use.

Neutralising solution to stop BAC biocidal effects had the following composition per litre: 10 ml of a 34 g/l KH₂PO₄ solution adjusted to pH = 7.2 with NaOH_(aq), 3 g soy lecithin, 5 g Na₂S₂O₃, 1 g L-histidine, 30 ml Tween 80 and adjusted with distilled water. This solution was autoclaved at 121 °C for 20 min and kept at 4 °C until use.

2.6. Experimental design

A two-phase experimental procedure was designed to evaluate the tolerance development (TD) in dual-species *L. monocytogenes*-*E. coli* biofilms to the application of PRN-BAC combined treatments.

2.6.1. Phase 1: sublethal exposures

In this phase, 24 h *L. monocytogenes*-*E. coli* biofilms were exposed to a different number of consecutive PRN-BAC sublethal exposures with or without recovery periods as schematised in Fig. 1. Sublethal concentrations were determined in previous assays (data not shown). Both PRN and BAC stock solutions were diluted in mTSB at concentrations for sublethal exposure (Fig. 1). Additionally, a negative control series in which PRN and BAC solutions used were substituted by equal volumes of sterile deionised water was included in each experiment. Thus, the experimental approaches and concentrations used were as follows:

2.6.1.1. Experimental approach 1: Short term PRN-BAC exposure.

In this approach, samples were subjected to two consecutive PRN-BAC exposures. At 24 h, the bulk phase of the samples was carefully pipetted out and wells were re-filled with 1 ml mTSB containing PRN to a final concentration of 50 μg/ml PRN for 1 h contact time without agitation at room temperature. Then, 1.5 ml mTSB containing BAC to a final concentration of 25 μg/ml was poured into each well and plates were placed back at 25°C/100 rpm. At 48 h, bulk phase was newly removed and biofilms were exposed to 1 ml mTSB containing PRN to a final concentration of 100 μg/ml PRN, let to dwell 1 h statically at room temperature. Then, 1.5 ml mTSB containing BAC to a final concentration of 50 μg/ml was added to each sample and plates were placed back at 25°C/100 rpm until next step. Negative controls were subjected to the same exposure scheme but the volumes of PRN and BAC used were substituted by equal volumes of sterile deionised water (Fig. 1).

In this approach, TD quantification carried out as described below, was determined at 72 h (Fig. 1).

2.6.1.2. Experimental approach 2: Short term PRN-BAC exposure with recovery periods.

In this approach, exposure to treatments was carried out as described for approach 1 but after every PRN-BAC

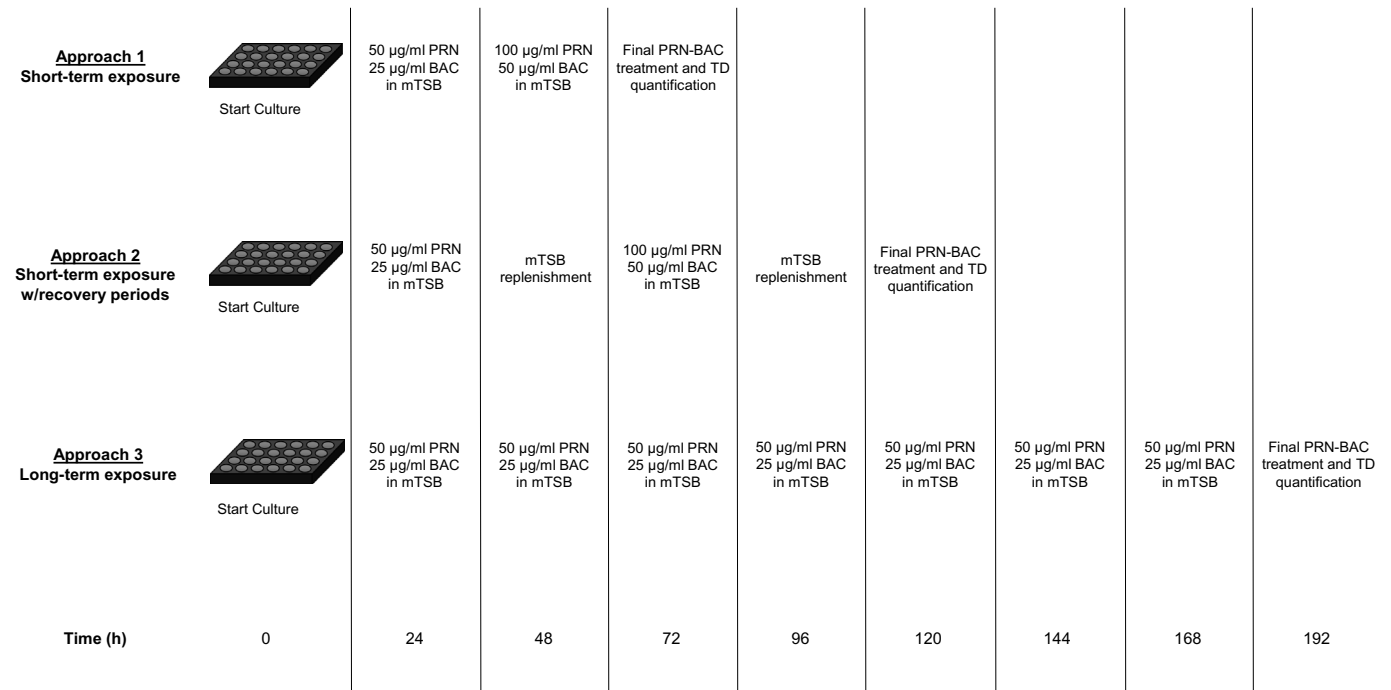


Fig. 1. Experimental approaches followed in exposed samples. At each experimental time, bulk phase was removed and solutions of pronase (PRN) and benzalkonium chloride (BAC) were applied at room temperature at the indicated concentrations. For final PRN-BAC treatment a 100 μg/ml PRN solution in 0.1 mM Tris-HCl followed by the addition of a 50 μg/ml BAC solution was used and tolerance development (TD) quantification was performed as described in the text. In all cases, control samples were cultured and treated the same way substituting PRN and BAC solutions by equal volumes of sterile deionised water. (See text for further details).

exposure the bulk phase was pipetted out and replenished by 1 ml plain mTSB (i.e. at 48 and 96 h), and incubated at 25°C/100 rpm for 24 h (Fig. 1). As in the previous approach, controls were subjected to the same exposure scheme but PRN and BAC solutions used were again substituted by equal volumes of sterile deionised water (Fig. 1).

In this second case, TD quantification to PRN-BAC as described below was carried out at 120 h (Fig. 1).

2.6.1.3. Experimental approach 3: Long term PRN-BAC exposure. In this third approach, bulk phases of the samples were pipetted out and wells were re-filled with 1 ml mTSB containing PRN to a final concentration of 50 µg/ml followed by the addition of 1.5 ml mTSB containing BAC to a final concentration of 25 µg/ml at 24, 48, 72, 96 and 168 h (Fig. 1). As performed in the previous approaches, for negative controls PRN and BAC volumes used were substituted by equal volumes of sterile deionised water (Fig. 1).

At 190 h, final treatment and quantification of the TD to PRN-BAC, as described below, was carried out (Fig. 1).

2.6.2. Phase 2: Quantification of tolerance development (TD) in *L. monocytogenes*-*E. coli* biofilms to PRN-BAC combined treatments

After growth, exposed and control dual-species biofilms were evaluated by assessing the state of the biofilm and the TD after a final exposure to a PRN-BAC treatment.

Biofilm condition was assessed by quantifying the number of AVC and the OA by undamaged (green) cells on the SS coupons in both experimental series (see sections 2.3. and 2.4., respectively). Results were expressed in log CFU/cm² and in mm² for AVC and OA, respectively.

TD was quantified by comparing the AVC and OA values of the dual-species samples previously exposed to sublethal PRN-BAC concentrations with those obtained in controls after a final PRN-BAC treatment. Specifically, 3 new coupons per series were transferred to a new well and treated again with 1 ml of a 100 µg/ml PRN solution in 0.1 mM Tris-HCl for 1 h contact time at room temperature without agitation. Then, 1.5 ml of a 50 µg/ml BAC solution in deionised water was added and let to dwell statically for 10 min at room temperature. To stop BAC effects, samples were newly transferred to a well containing 1 ml of sterile neutraliser and immersed for 10 s at room temperature. After neutralisation, coupons were processed for AVC and microscopy assays as described above (sections 2.3. and 2.4.).

2.7. Statistical analysis

An independent-samples Student's *t*-test was performed to determine differences between values using Microsoft Excel 2016. Significance was expressed at the 95% confidence level ($\alpha = 0.05$) or greater.

3. Results

3.1. Approach 1: short term exposure

Comparison between the condition of *L. monocytogenes*-*E. coli* dual-species biofilm formed without any exposure to PRN-BAC and after two consecutive exposures to double-fold increasing PRN-BAC concentrations was carried out according to experimental approach 1 (Fig. 1). Obtained results showed that the number of *L. monocytogenes* adhered in exposed biofilms was significantly lower comparing with that obtained in the controls (3.95 ± 0.48 and 6.37 ± 0.46 log CFU/cm², respectively) (Fig. 2A). No significant differences were obtained in the number of *E. coli* in unexposed and exposed biofilms (Fig. 2A).

The final treatment with PRN-BAC decreased the number of cells in all experimental series assayed about 1.5–2.0 log CFU/cm² (Fig. 2B). However, when the resistance of exposed and control biofilms was compared in terms of log reduction of the number of adhered cells of both species, no significant differences ($P > 0.05$) were obtained (Fig. 3). Taken together, these results seem to indicate that *L. monocytogenes*-*E. coli* dual-species biofilms did not acquire any tolerance to PRN-BAC treatment after exposure following approach 1.

Microscopic images showed no significance in OA values by undamaged (green) cells between control biofilms (0.84 ± 0.06 mm²) compared to PRN-BAC exposed samples (0.70 ± 0.16 mm²) (Fig. 4). Last PRN-BAC treatment applied to the samples obtained with approach 1 had visible effects on control and exposed biofilms regarding the 2D morphology of the biofilm, however, no significant differences were observed in terms of OA values (Fig. 4).

3.2. Approach 2: short term exposure with recovery periods

This second approach was specifically designed to check whether *L. monocytogenes*-*E. coli* dual-species biofilms were able to develop tolerance to PRN-BAC treatments when incorporating 24 h-recovery periods after each of the 2 exposures scheduled in the approach 1 using fresh mTSB (Fig. 1).

No significant differences ($P > 0.05$) were obtained in AVC values of control samples neither in *L. monocytogenes* nor in *E. coli* (6.87 ± 0.08 and 7.72 ± 0.30 log CFU/cm², respectively) compared to those obtained in controls of approach 1 (Fig. 2A). This indicated that the inclusion of recovery periods in between exposures did not produce a higher number of cells to be attached to the surface. On the other hand, AVC outcomes in exposed samples were lower in both species compared to those obtained in approach 1 (3.11 ± 0.16 CFU/cm² for *L. monocytogenes* and 6.53 ± 0.18 CFU/cm² for *E. coli*) (Fig. 2A).

Tolerance development to PRN-BAC of the dual-species biofilms of approach 2 was then assessed in terms of log reduction and compared to results obtained in approach 1. Outcomes demonstrated that the recovery periods included caused a drop in the log reduction values in both unexposed (control) and exposed samples (Fig. 3). Considering that AVC values before final PRN-BAC treatment between approaches 1 and 2 were not significantly different in neither species (Fig. 2), that meant that the overall resistance of the cells of *L. monocytogenes*-*E. coli* biofilms to the PRN-BAC treatment was higher, even in the case of controls (Fig. 3). Indeed, whereas the log reduction values obtained in approach 1 ranged from 1.56 ± 0.25 to 2.34 ± 0.50 log CFU/cm², using this second exposure schedule the variation among log reduction outcomes was from 0.13 ± 0.08 to 1.06 ± 0.23 log CFU/cm² (Fig. 3). This increased resistance was observed in both species, even though *L. monocytogenes* presented bigger differences than *E. coli* if values between approaches are compared (Fig. 3). Surprisingly, a higher resistance to PRN-BAC treatments was also observed in control biofilms respecting to those values obtained in exposed biofilms of approach 1 (Fig. 3).

3.3. Approach 3: long term exposure

In order to check if longer incubation times with the sublethal treatments would influence the final resistance of the biofilm, a third experimental approach consisting in five consecutive exposures to sublethal PRN-BAC concentrations was used (Fig. 1).

Before final PRN-BAC treatment, control samples did not present significance in *E. coli* (7.85 ± 0.26 log CFU/cm²) respecting to previous approaches whereas in *L. monocytogenes* the AVC value was

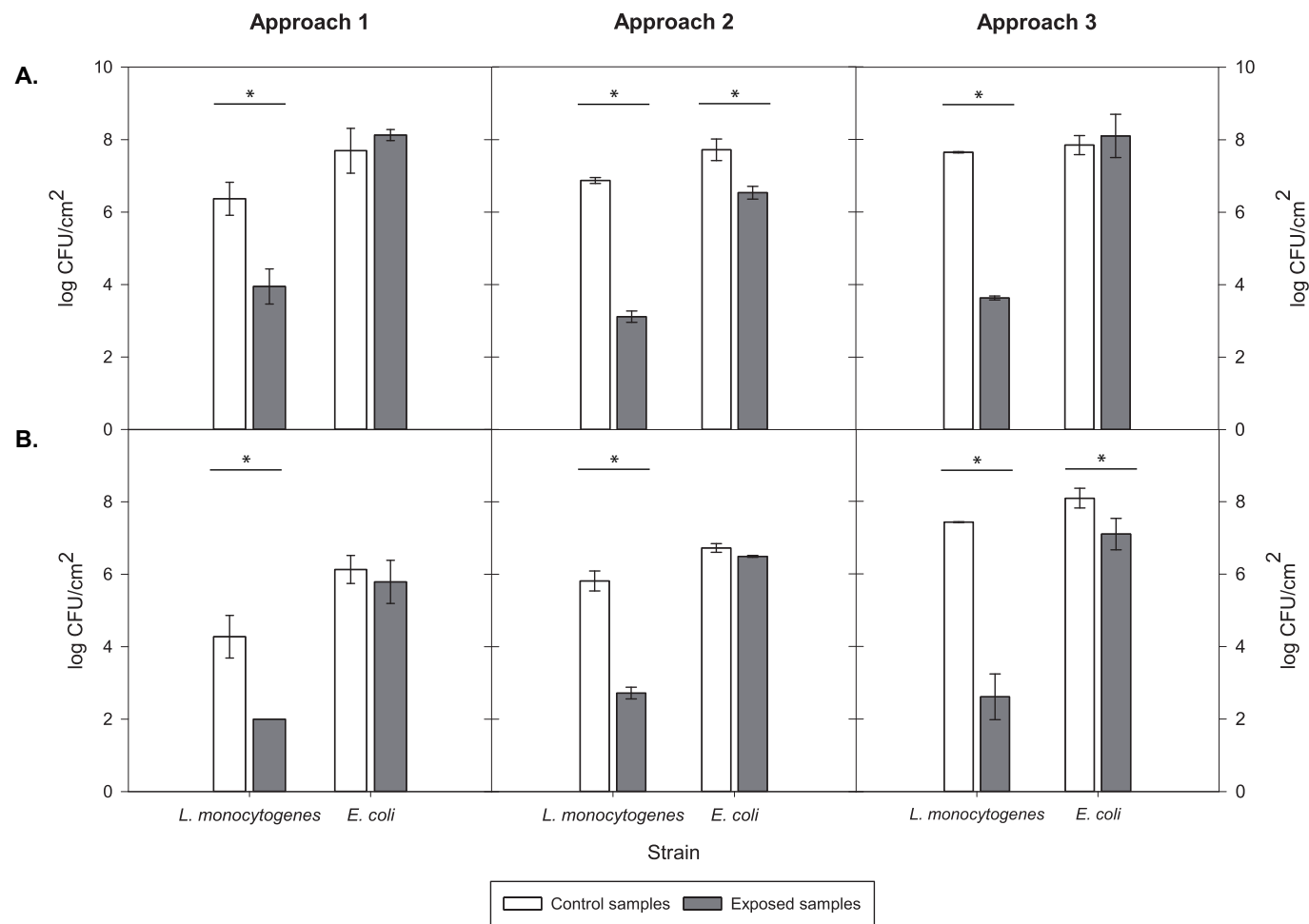


Fig. 2. Adhered viable culturable cells (AVC) counts obtained in all experimental approaches A) before final and B) after final 100 µg/ml pronase followed by 50 µg/ml benzalkonium chloride combined treatment. Exposed samples refer to those subjected to a previous PRN-BAC exposure whereas in control samples biocides solutions volumes were replaced with water (see section 2.6. for further detail). Asterisks indicate statistical significance ($P < 0.05$).

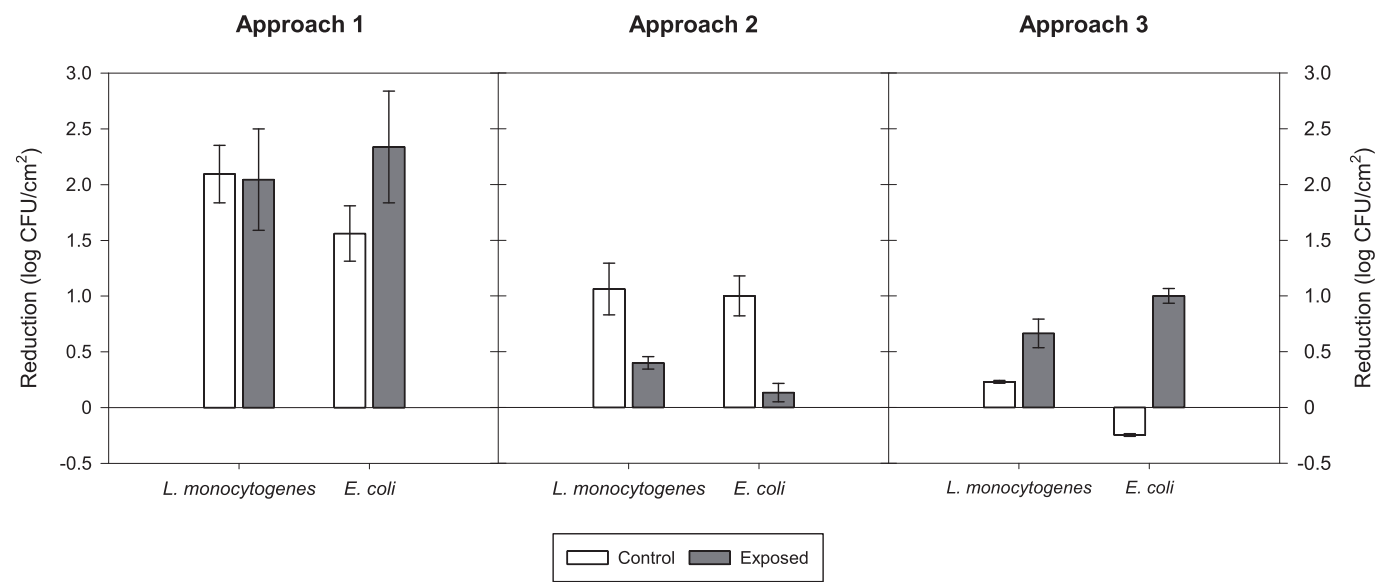


Fig. 3. Log reductions obtained in all experimental approaches after the application of a final 100 µg/ml pronase (PRN) followed by 50 µg/ml benzalkonium chloride combined (BAC) treatment. Exposed samples refer to those subjected to a previous PRN-BAC exposure whereas in control samples biocides solutions volumes were replaced with water (see section 2.6. for further detail). In approach 3, the negative reduction in *E. coli* refers to a slight increase in the number of adhered viable cultivable cells recovered after final PRN-BAC treatment. Asterisks indicate statistical significance ($P < 0.05$).

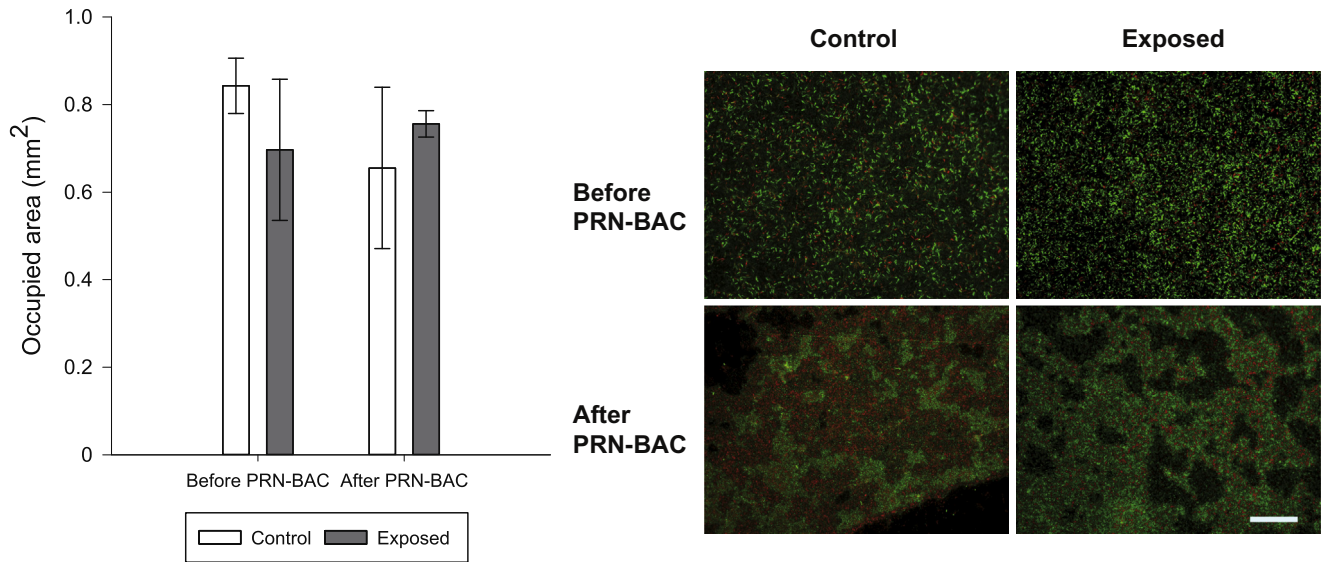


Fig. 4. Values of occupied area obtained after analysis of 25-field mosaics of approach 1 before and after the application of pronase (PRN)-benzalkonium chloride (BAC) final treatment. Exposed samples were subjected to a previous PRN-BAC exposure whereas control samples biocide solution volumes were replaced with water (see section 2.6.1.1. for further detail). Scale bar = 50 µm.

significantly higher (7.65 ± 0.02 log CFU/cm²) compared to AVC outcomes obtained in previous approaches (Fig. 2A). In exposed biofilms, AVC values were significantly ($P < 0.05$) higher than those of approach 2 (3.63 ± 0.05 and 8.10 ± 0.59 log CFU/cm² for *L. monocytogenes* and *E. coli*, respectively) (Fig. 2A).

TD quantification demonstrated the lowest log reductions among control samples in *L. monocytogenes* (0.23 ± 0.12 log CFU/cm²) even presenting negative values (i.e. a slight AVC increase) in *E. coli* (-0.25 ± 0.01 log CFU/cm²) (Fig. 3). Among exposed samples, the log reductions were higher than those of approach 2 but significantly lower than those of approach 1 (Fig. 3), with final AVC values of 2.61 ± 0.63 and 7.10 ± 0.43 log CFU/cm² in *L. monocytogenes* and *E. coli*, respectively (Fig. 2).

Analysis of microscopy images for OA determination before final PRN-BAC treatment, showed no significance between series

yielding values of 0.87 ± 0.18 and 0.77 ± 0.12 mm² for control and exposed samples, respectively (Fig. 5). Besides, images corresponding to control samples presented a cloudy-like structures stained in green that were not visible among the exposed biofilms. After final PRN-BAC treatment, OA value in exposed samples (0.54 ± 0.09 mm²) was significantly lower than controls (0.75 ± 0.25 mm²). This effect was also visible among images where a lower green signal was obtained whereas the proportion of red-stained cells rised (Fig. 5).

4. Discussion

In the present study, the capability of *L. monocytogenes*-*E. coli* biofilms to grow and develop tolerance to PRN-BAC combined treatments after sublethal exposure following three different

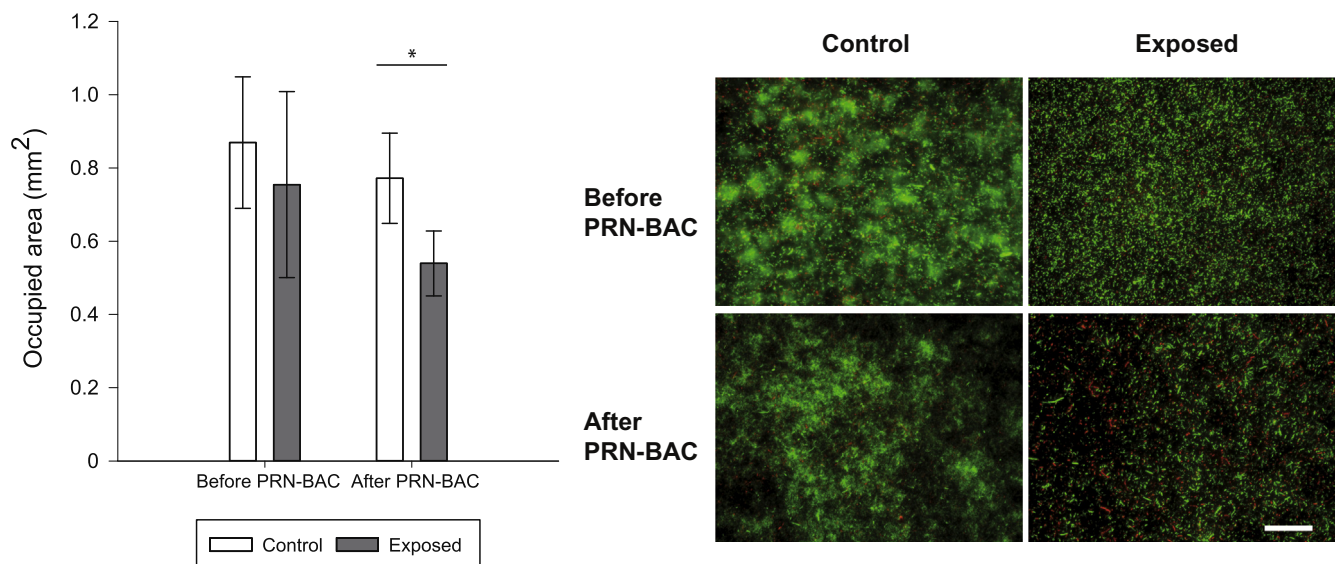


Fig. 5. Values of occupied area obtained after analysis of 25-field mosaics of approach 3 before and after the application of pronase (PRN)-benzalkonium chloride (BAC) final treatment. Exposed samples were subjected to a previous PRN-BAC exposure whereas control samples biocide solution volumes were replaced with water (see section 2.6.1.3. for further detail). Asterisk indicates statistical significance ($P < 0.05$). Scale bar = 50 µm.

approaches was determined. Each individual approach was intended to assess the effects after three feasible situations related to the sanitation practise in the food industry. Thus, approaches 1 and 2 simulated procedures in which treatments are applied using increasing concentrations daily or every other day for biofilm removal. On the other hand, approach 3 intended to emulate the effects of one-week daily cleaning and disinfection routines using the same concentrations of PRN and BAC.

Before any antimicrobial treatment can be applied, its efficacy against a particular target must be previously tested. Despite this, to date, no standardised methods for antibiofilm testing are available (Malone et al., 2016) and none of them considers TD as a parameter, basic to determine the appropriate dosage to apply for proper cleaning and disinfection, avoiding the generation of antimicrobial tolerances.

Results showed that the number of *L. monocytogenes* AVC decreased after sublethal exposure to PRN-BAC regardless of the exposure scheme used (Fig. 2A). Contrarily, in *E. coli* the presence of PRN-BAC did not affect the number of AVC on SS coupons in approaches 1 and 3 (Fig. 2A). In approach 2, the number of AVC of *E. coli* in exposed samples was slightly lower compared to control samples but the differences between series were smaller than those obtained in *L. monocytogenes* (Fig. 2A). These results were not surprising considering that Gram-negatives are generally more resistant to QACs than Gram-positives (McDonnell and Russell, 1999) and were in accordance with results obtained by Machado et al. (2012) who observed that the number of adhered cells in 6-day old dual-species biofilms of *Pseudomonas aeruginosa*-*E. coli* was not affected by presence of 328.5 µg/ml BAC in the culture medium.

Obtained results showed that the acquired tolerance to PRN-BAC by *L. monocytogenes*-*E. coli* dual-species biofilms is the result of the following two interdependent factors: i) the effects produced by the cultural features (i. e. total incubation time and the way in which biofilms were exposed to PRN and BAC) of each experimental approach giving rise to a different *L. monocytogenes*-*E. coli* dual-species biofilms, and ii) the specific physiological effects provoked by the sublethal PRN-BAC exposures.

Regarding the first comparison, significant higher values of log reduction were obtained in the dual-species biofilms subjected to short-term approach (approach 1) compared to those obtained when short-term with recovery periods and long-term exposures (approaches 2 and 3, respectively) were applied (Fig. 3). In other words, biofilms grown in approach 1 were much less tolerant to PRN-BAC. This fact, together with the absence of significance between AVC values in all the experimental approaches before and after the application of the last PRN-BAC treatment, pointed out that longer incubation times (approaches 2 and 3) and the incorporation of recovery periods (approach 2) gave rise to structures with higher PRN-BAC tolerance.

The TD in terms of the number of AVC that resist the last PRN-BAC treatment after sublethal exposure was only observed in approach 2, in which significant differences ($P < 0.05$) in the log reduction between control and exposed dual-species biofilms were observed in both strains.

Bacterial recovery after sublethal injured has been reported to contribute to adaptation and hardening of cells after stresses (Wesche et al., 2009). Among foodborne pathogens, this recovery period is rather short; around 1–5 h at 25–37 °C in rich broth (Wu, 2008). Thus, in biofilms, these factors can promote the appearance of persistent strains in food-related environments (Simões et al., 2006). Considering this, it is logical to think that the incorporation of recovery periods in approach 2, allowed injured bacteria to repair damaged structures, and develop stress-induced strategies to prepare the cells for further external aggressions.

Membrane alterations together with the expression of BAC-induced efflux pumps, have been previously reported by several authors as the main cause of BAC-sensitivity reduction both in *L. monocytogenes* (Aase et al., 2000; Bisbiroulas et al., 2011; Romanova et al., 2006; To et al., 2002) and *E. coli* (Bore et al., 2007; Langsrud et al., 2004; Moen et al., 2012). Following these ideas, the recovery periods in approach 2 could have permitted the surviving subpopulation after PRN-BAC exposure to undergo mechanisms for membrane damage repair, as previously observed in *L. monocytogenes* after the application of sanitisers (Donnelly, 2002).

Additionally, these periods could have allowed cells in the bulk phase to have enough time to repair possible damaged cellular structures and express chromosome and plasmid-encoded efflux pumps, therefore causing the extrusion of BAC outside of the cell. With this regard, Tamburro et al. (2015) observed an alteration in the gene expression pattern concomitant with a higher tolerance to BAC in *L. monocytogenes* after exposure to 10 µg/ml BAC, highlighting a significant increase in the expression of *mdrL* (efflux pump) and *sigB* (transcription factor) genes. Similar phenomena have also been reported in *E. coli* after BAC sublethal exposure (Bore et al., 2007). This increased tolerance is even more relevant if we consider that these pumps can be effective against other molecules therefore promoting BAC-induced multidrug resistance (Aase et al., 2000; Rakic-Martinez et al., 2011).

Furthermore, the recovery periods in approach 2 could have permitted to the exposed biofilm samples to undertake species spatial reorganisation after PRN-BAC contact and that this increased the overall protection of the structure. With this regard, previous studies published point out that species cross-protective effects depending on spatial distribution in mixed biofilms after antimicrobial exposure, significantly affects the overall biofilm tolerance (Lee et al., 2014; Leriche et al., 2003). Such cross-protection would be of the weakest by the most resistant species as previously stated (Lee et al., 2014). So, if this was the case, *L. monocytogenes* would be protected by *E. coli* since, generally, the latter is more resistant to BAC than Gram-positives (McDonnell and Russell, 1999).

Nevertheless, the specific effect of the recovery periods has not been previously described in biofilms. In planktonic cells of *Pseudomonas* spp., it was previously observed that the adaptive resistance acquired after short term exposure to 200 µg/ml of BAC was rapidly lost after overnight incubation in absence of the disinfectant (Langsrud and Sundheim, 1997). Interestingly, the authors also demonstrated that the level of tolerance of *Pseudomonas* spp. to the application of BAC and BAC-EDTA was conditioned by the treatment applied during the sublethal exposure.

Besides, sublethal exposures to PRN-BAC would probably determine important specific biological modifications in the final *L. monocytogenes*-*E. coli* dual-species biofilm. In fact, previous studies have demonstrated that the presence of BAC during biofilm formation by *E. coli* gives rise to a denser matrix richer in proteins and polysaccharides (Machado et al., 2012). In this latter case, the possible presence of high amounts of protective colanic acid would make the matrix barrier to be very difficult to penetrate (Flemming and Wingender, 2010; Sutherland, 2001).

The way in which all these biological processes takes place in industrial settings, is directly related with the way in which sanitation procedures are applied. Insufficient rinsing is one of the most common causes of presence of sublethal amounts of biocides among surfaces treated with QACs in the food industry (Møretro et al., 2017). This fact together with the time in-between scheduled cleaning and disinfection protocols applied in a processing plant, would permit bacteria to recover from biocide injury eventually boosting their resistance to chemicals as demonstrated by

the results obtained.

5. Conclusions

Outcomes in this work have demonstrated that the manner in which *L. monocytogenes*-*E. coli* biofilms are exposed to sublethal concentrations of PRN-BAC influences the subsequent resistance to this combined treatment. Firstly, it was observed that recovery periods lead to a selection of a resistant subpopulation compared to unexposed samples. In addition to this, it was also observed that longer incubation times also influenced the resistance of the biofilm. Therefore, for biofilm eradication, a thorough optimisation not only of the right amounts of antimicrobial compounds utilised but also a proper time scheduling would be necessary prior to the application of any sanitation procedure in order to obtain proper bactericidal effects while avoiding the selection of resistant variants.

Acknowledgments

This research was financially supported by the Spanish Ministerio de Economía y Competitividad (ENZYMONO, AGL2010-22212-C02-02). P. Rodríguez-López acknowledges the financial support from the FPI programme (Grant number: BES-2011-050544). All authors acknowledge S. Rodríguez and T. Blanco (laboratory technicians, IIM-CSIC) for their help throughout all the study.

All results and conclusions presented in this article are part of P. Rodríguez-López PhD dissertation in the framework of the microbiology doctorate programme of the Autonomous University of Barcelona.

References

- Aase, B., Sundheim, G., Langsrud, S., Rørvik, L.M., 2000. Occurrence of and a possible mechanism for resistance to a quaternary ammonium compound in *Listeria monocytogenes*. *Int. J. Food Microbiol.* 62, 57–63. [http://dx.doi.org/10.1016/S0168-1605\(00\)00357-3](http://dx.doi.org/10.1016/S0168-1605(00)00357-3).
- Abdallah, M., Benoliel, C., Drider, D., Dhulster, P., Chihib, N.E., 2014. Biofilm formation and persistence on abiotic surfaces in the context of food and medical environments. *Arch. Microbiol.* 196, 453–472. <http://dx.doi.org/10.1007/s00203-014-0983-1>.
- Allison, D.G., 2003. The biofilm matrix. *Biofouling* 19, 139–150. <http://dx.doi.org/10.1080/0892701031000072190>.
- Augustin, M., Ali-Vehmas, T., Atroshi, F., 2004. Assessment of enzymatic cleaning agents and disinfectants against bacterial biofilms. *J. Pharm. Pharm. Sci.* 7, 55–64.
- Bisbiroulas, P., Psylou, M., Iliopoulou, I., Diakogiannis, I., Berberi, A., Mastroicis, S.K., 2011. Adaptational changes in cellular phospholipids and fatty acid composition of the food pathogen *Listeria monocytogenes* as a stress response to disinfectant sanitizer benzalkonium chloride. *Lett. Appl. Microbiol.* 52, 275–280. <http://dx.doi.org/10.1111/j.1472-765X.2010.02995.x>.
- Bore, E., Hébraud, M., Chafsey, I., Chambon, C., Skjæret, C., Moen, B., Mørseth, T., Langsrud, Ø., Rudi, K., Langsrud, S., 2007. Adapted tolerance to benzalkonium chloride in *Escherichia coli* K-12 studied by transcriptome and proteome analyses. *Microbiology* 153, 935–946. <http://dx.doi.org/10.1099/mic.0.29288-0>.
- Donnelly, C.W., 2002. Detection and isolation of *Listeria monocytogenes* from food samples: implications of sublethal injury. *J. AOAC Int.* 85, 495–500.
- EFSA, 2015. The European Union summary report on trends and sources of zoonoses, zoonotic agents and food-borne outbreaks in 2014. *EFSA J.* 13, 4329. <http://dx.doi.org/10.2903/j.efsa.2015.4329>.
- Flemming, H.C., Neu, T.R., Wozniak, D.J., 2007. The EPS matrix: the “House of Biofilm Cells”. *J. Bacteriol.* 189, 7945–7947. <http://dx.doi.org/10.1128/JB.00858-07>.
- Flemming, H.C., Wingender, J., 2010. The biofilm matrix. *Nat. Rev. Microbiol.* 8, 623–633. <http://dx.doi.org/10.1038/nrmicro2415>.
- Holoh, J.T., Taylor, J.H., Dawson, D.J., Hall, K.E., 2002. Biocide use in the food industry and the disinfectant resistance of persistent strains of *Listeria monocytogenes* and *Escherichia coli*. *J. Appl. Microbiol.* 92, 1115–1205. <http://dx.doi.org/10.1046/j.1365-2672.92.5s1.18.x>.
- Jiang, X., Yu, T., Liang, Y., Ji, S., Guo, X., Ma, J., Zhou, L., 2016. Efflux pump-mediated benzalkonium chloride resistance in *Listeria monocytogenes* isolated from retail food. *Int. J. Food Microbiol.* 217, 141–145. <http://dx.doi.org/10.1016/j.jifoodmicro.2015.10.022>.
- Langsrud, S., Sundheim, G., 1997. Factors contributing to the survival of poultry associated *Pseudomonas* spp. exposed to a quaternary ammonium compound. *J. Appl. Microbiol.* 82, 705–712. <http://dx.doi.org/10.1046/j.1365-2672.1997.00186.x>.
- Langsrud, S., Sundheim, G., Holck, A.L., 2004. Cross-resistance to antibiotics of *Escherichia coli* adapted to benzalkonium chloride or exposed to stress-inducers. *J. Appl. Microbiol.* 96, 201–208. <http://dx.doi.org/10.1046/j.1365-2672.2003.02140.x>.
- Lee, K.W.K., Periasamy, S., Mukherjee, M., Xie, C., Kjelleberg, S., Rice, S.A., 2014. Biofilm development and enhanced stress resistance of a model, mixed-species community biofilm. *ISME J.* 8, 894–907. <http://dx.doi.org/10.1038/ismej.2013.194>.
- Leriche, V., Briand, R., Carpentier, B., 2003. Ecology of mixed biofilms subjected daily to a chlorinated alkaline solution: spatial distribution of bacterial species suggests a protective effect of one species to another. *Environ. Microbiol.* 5, 64–71. <http://dx.doi.org/10.1046/j.1462-2920.2003.00394.x>.
- Machado, I., Lopes, S.P., Sousa, A.M., Pereira, M.O., 2012. Adaptive response of single and binary *Pseudomonas aeruginosa* and *Escherichia coli* biofilms to benzalkonium chloride. *J. Basic Microbiol.* 52, 43–52. <http://dx.doi.org/10.1002/jobm.201100137>.
- Malone, M., Goeres, D.M., Gosbell, I., Vickery, K., Jensen, S., Stoodley, P., 2016. Approaches to biofilm-associated infections: the need for standardized and relevant biofilm methods for clinical applications. *Expert Rev. Anti. Infect. Ther.* <http://dx.doi.org/10.1080/14787210.2017.1262257>.
- McDonnell, G., Russell, A.D., 1999. Antiseptics and disinfectants: activity, action, and resistance. *Clin. Microbiol. Rev.* 12, 147–179. <http://dx.doi.org/10.1007/s13398-014-0173-7>.
- Meireles, A., Borges, E., Gaiouris, E., Simões, M., 2016. The current knowledge on the application of anti-biofilm enzymes in the food industry. *Food Res. Int.* 86, 140–146. <http://dx.doi.org/10.1016/j.foodres.2016.06.006>.
- Moen, B., Rudi, K., Bore, E., Langsrud, S., 2012. Subminimal inhibitory concentrations of the disinfectant benzalkonium chloride select for a tolerant subpopulation of *Escherichia coli* with inheritable characteristics. *Int. J. Mol. Sci.* 13, 4101–4123. <http://dx.doi.org/10.3390/ijms13044101>.
- Molobela, I.P., Cloete, T.E., Beukes, M., 2010. Protease and amylase enzymes for biofilm removal and degradation of extracellular polymeric substances (EPS) produced by *Pseudomonas fluorescens* bacteria. *Afr. J. Microbiol. Res.* 4, 1515–1524.
- Mørseth, T., Schirmer, B.C.T., Heir, E., Fagerlund, A., Hjemli, P., Langsrud, S., 2017. Tolerance to quaternary ammonium compound disinfectants may enhance growth of *Listeria monocytogenes* in the food industry. *Int. J. Food Microbiol.* 241, 215–224. <http://dx.doi.org/10.1016/j.jifoodmicro.2016.10.025>.
- Orgaz, B., Neufeld, R.J., SanJose, C., 2007. Single-step biofilm removal with delayed release encapsulated Pronase mixed with soluble enzymes. *Enzyme Microb. Technol.* 40, 1045–1051. <http://dx.doi.org/10.1016/j.enzmictec.2006.08.003>.
- Rakic-Martinez, M., Drevets, D.A., Dutta, V., Katic, V., Kathariou, S., 2011. *Listeria monocytogenes* strains selected on ciprofloxacin or the disinfectant benzalkonium chloride exhibit reduced susceptibility to ciprofloxacin, gentamicin, benzalkonium chloride, and other toxic compounds. *Appl. Environ. Microbiol.* 77, 8714–8721. <http://dx.doi.org/10.1128/AEM.05941-11>.
- Rodríguez-López, P., Carballo-Justo, A., Draper, L.A., Cabo, M.L., 2017. Removal of *Listeria monocytogenes* dual-species biofilms using enzyme-benzalkonium chloride combined treatments. *Biofouling* 33, 45–58. <http://dx.doi.org/10.1080/08927014.2016.1261847>.
- Rodríguez-López, P., Saá-Ibáñez, P., Mosquera-Fernández, M., López-Cabo, M., 2015. *Listeria monocytogenes* – carrying consortia in food industry. Composition, subtyping and numerical characterisation of mono-species biofilm dynamics on stainless steel. *Int. J. Food Microbiol.* 206, 84–95. <http://dx.doi.org/10.1016/j.jifoodmicro.2015.05.003>.
- Romanova, N.A., Wolffs, P.F.G., Brovko, L.Y., 2006. Role of efflux pumps in adaptation and resistance of *Listeria monocytogenes* to benzalkonium chloride role of efflux pumps in adaptation and resistance of *Listeria monocytogenes* to benzalkonium chloride. *Appl. Environ. Microbiol.* 72, 3498–3503. <http://dx.doi.org/10.1128/AEM.72.5.3498-3503.2006>.
- Saá-Ibáñez, P., Herrera, J.J.R., Vázquez-Sánchez, D., Cabo, M.L., 2012a. Adherence kinetics, resistance to benzalkonium chloride and microscopic analysis of mixed biofilms formed by *Listeria monocytogenes* and *Pseudomonas putida*. *Food Control* 25, 202–210. <http://dx.doi.org/10.1016/j.foodcont.2011.10.002>.
- Saá-Ibáñez, P., Herrera, J.J.R., Vázquez-Sánchez, D., Parada, A., Cabo, M.L., 2012b. A new and efficient method to obtain benzalkonium chloride adapted cells of *Listeria monocytogenes*. *J. Microbiol. Methods* 91, 57–61. <http://dx.doi.org/10.1016/j.mimet.2012.07.009>.
- Simões, M., Simões, L.C., Machado, I., Pereira, M.O., Vieira, M.J., 2006. Control of flow-generated biofilms with surfactants. *Food Bioprod. Process* 84, 338–345. <http://dx.doi.org/10.1205/fbp06022>.
- Simões, M., Simões, L.C., Vieira, M.J., 2010. A review of current and emergent biofilm control strategies. *LWT - Food Sci. Technol.* 43, 573–583. <http://dx.doi.org/10.1016/j.lwt.2009.12.008>.
- Sutherland, I.W., 2001. Biofilm exopolysaccharides: a strong and sticky framework. *Microbiology* 147, 3–9. <http://dx.doi.org/10.1051/laite:2001108>.
- Sutton, S., 2011. Accuracy of plate counts. *J. Valid. Technol.* 17, 42–46.
- Tamburro, M., Ripabelli, G., Vitullo, M., Dallman, T.J., Pontello, M., Amar, C.F.L., Sammarco, M.L., 2015. Gene expression in *Listeria monocytogenes* exposed to sublethal concentration of benzalkonium chloride. *Comp. Immunol. Microbiol. Infect. Dis.* 40, 31–39. <http://dx.doi.org/10.1016/j.cimid.2015.03.004>.
- To, M.S., Favrin, S., Romanova, N., Griffiths, M.W., 2002. Postadapational resistance to benzalkonium chloride and subsequent physicochemical modifications of *Listeria monocytogenes*. *Appl. Environ. Microbiol.* 68, 5258–5264. <http://dx.doi.org/10.1046/j.1365-2672.92.5s1.18.x>.

- dx.doi.org/10.1128/AEM.68.11.5258-5264.2002.
- van der Veen, S., Abee, T., 2011. Mixed species biofilms of *Listeria monocytogenes* and *Lactobacillus plantarum* show enhanced resistance to benzalkonium chloride and peracetic acid. *Int. J. Food Microbiol.* 144, 421–431. <http://dx.doi.org/10.1016/j.ijfoodmicro.2010.10.029>.
- Wesche, A.M., Gurtler, J.B., Marks, B.P., Ryser, E.T., 2009. Stress, sublethal injury, resuscitation, and virulence of bacterial foodborne pathogens. *J. Food Prot.* 72, 1121–1138.
- Wu, V.C.H., 2008. A review of microbial injury and recovery methods in food. *Food Microbiol.* 25, 735–744. <http://dx.doi.org/10.1016/j.fm.2008.04.011>.